

Experiment: File Integrity Verification using SHA-256 Hashing

Aim

To verify the integrity of a copied file by generating and comparing SHA-256 hash values using Python.

Algorithm

1. Read the source file path and destination folder path from the user.
2. Check whether the source file and destination folder exist.
3. Copy the source file to the destination folder.
4. Generate the SHA-256 hash values of both the original and copied files.
5. Compare the hash values and display whether the file integrity is verified or failed.

Python Code

```
import hashlib
```

```
import shutil
```

```
import os
```

```
source = input("Source File: ").strip().strip('"')
```

```
destination_folder = input("Destination Folder: ").strip().strip('"')
```

```
if not os.path.isfile(source):
```

```
    print("Source file not found!")
```

```
    exit()
```

```
if not os.path.isdir(destination_folder):
```

```
print("Destination folder not found!")

exit()

filename = os.path.basename(source)
destination = os.path.join(destination_folder, filename)

shutil.copyfile(source, destination)

def get_hash(file):
    sha = hashlib.sha256()
    with open(file, "rb") as f:
        while True:
            data = f.read(4096)
            if not data:
                break
            sha.update(data)
    return sha.hexdigest()

h1 = get_hash(source)
h2 = get_hash(destination)

print("\nOriginal Hash :", h1)
print("Copied Hash  :", h2)

if h1 == h2:
```

```
print("\nIntegrity Verified")  
  
else:  
  
    print("\nIntegrity Failed")
```

Sample Input

Source File:

C:\Users\saich\OneDrive\Desktop\Sai Charan Tej_Resume.pdf

Destination Folder:

C:\Users\saich\OneDrive\Desktop\digital lab

Sample Output

Original Hash :

269502b86e89a9d3ffdf6fa5eac127c5491400e16dfc656fe85c764647a3f228

Copied Hash :

269502b86e89a9d3ffdf6fa5eac127c5491400e16dfc656fe85c764647a3f228

Integrity Verified

Result

The program successfully copied the source file to the destination folder and generated SHA-256 hash values for both the original and copied files. Since both hash values were identical, the integrity of the copied file was successfully verified.